4. Personal Protective Equipment

Introduction

The hazards that are encountered in the workplace take many forms. Workers may be exposed to toxic gases, hazardous vapors, mists and fogs, hazardous materials in the form of particulate matter, hazardous liquids, fumes and smoke, as well as other physical hazards such as noise and electromagnetic radiation. Throughout this text, various methods have been discussed to protect the workers from exposures related to these materials. In this chapter, the use of personal protective equipment as a potential method for controlling exposure to hazardous substances will be discussed.

One of the first important questions that must be asked before determining the type of personal protective equipment to use is the form in which the hazardous material exists. This is important in determining the route of entry of the material into the body and, as a result, the type of protection that is required.

Toxic gases are most likely to enter the body by being inhaled. These gases attack the respiratory system and through the respiratory system can enter the bloodstream. Carbon monoxide, for example, enters the body in this manner. A secondary exposure from gases can occur when the gases act on the skin and eyes of the worker.

Vapors, mists, and fogs act in essentially the same manner as gases in the respiratory system. There is an additional danger as a result of condensation of the vapor, mist, or fog on the skin of the worker.

Particulate matter consists of solids suspended in the air. The very small particles may be inhaled and enter the respiratory system. These small particles can be passed into the bloodstream.

On the other hand, liquids, unless subject to vaporization, are more likely to attack the human body through the skin. Certain chemicals have a systemic action which, when entering the body through the skin, can cause harm throughout the entire system. Additionally, liquids can be ingested either accidentally or as a result of failure to clean thoroughly before smoking or eating.

Fumes and smoke may be inhaled and enter the respiratory tract or they may act on the skin. Fumes and smoke may also enter the eyes, causing damage.

Personal Protective Equipment as a Control

It should be noted that personal protective equipment is a last resort control method. Other controls should be attempted before using personal protective equipment. Can ventilation help solve the problem? Can a less hazardous material be substituted for a more hazardous one used in the process? Can engineering changes be made in the process and process equipment to eliminate the hazard? Can the worker be removed from contact with the source either by isolating the source or by isolating the worker? In those cases where the preferred methods of control do not provide adequate protection, then personal protective equipment should be prescribed until such time as permanent controls can be established.

Personal protective equipment provides a measure of control in the case of an emergency situation when a failure of the primary control method occurs. In such a case, personal protective equipment acts as emergency equipment. Personal protective equipment can also be used when no other control exists and the job is required to be performed by workers during the production process.

The basic categories of personal protective equipment that are available are:

1. Protection from inhalation of hazardous materials.
2. Protection from skin contact with hazardous materials.
3. Protection of the eyes from contact with hazardous materials.
4. Protection of the ears from hazardous exposure.
5. Protection from traumatic injury to a body part.
6. Protection from thermal stress.

Protection from Respirable Hazards

There are three basic types of personal protective equipment available for the protection of the worker from respirable hazards. These are:

1. Air purifying respirators
   - canister type
   - cartridge type
   - mechanical filter
2. Air supplied respirators
   - hose mask type
   - air-line respirator
   - abrasive blasting type respirator
3. Self-contained breathing type respirators
   - recirculating
   - open circulating

Before discussing each of these respirators in detail, let's look at the major criteria that should be considered when selecting the appropriate type of respirator.

The proper respirator must be used for a given situation. Otherwise, the respirator may provide a false sense of security and result in potential harm to the worker. The respirator used should be selected for the specific application involved.

There are many factors to consider in the selection of a respirator. The first question to consider is the hazardous material that is involved. What is its chemical makeup and its form? What are its characteristics? What are its chemical properties? What is the human physiological reaction to the material? What concentration of the material is present in the workplace?

Second, the conditions under which the worker will become exposed must be considered. Will the worker be required to perform normal work functions during exposure? This is likely to result when personal protective equipment is the final resort protection that is available. Or, is the respirator to be used only for emergency conditions; that is, when a failure of the existing controls or system occurs?

Closely related to the conditions in which the respirator will be used is the time or duration of the exposure. Will the respirator be required for quick escape as is the case in emergency use, or will the respirator be required for a longer period of time to perform work duties in the hazardous environment?

If the respirator is to be used during the performance of regular duties, it is important to determine the type of activity that must be performed while using the respirator. Will the work require much physical effort? If so, will the respirator inhibit the employee's ability to perform work in any way?

Another related question that must be asked but that does not directly involve the selection of the respirator is: "What is the need for additional protection?" While in the hazardous environment, will the employee be exposed in any way to the hazardous material? Is protection needed for the eyes or skin? Does the hazardous material act as a systemic agent through the exposed skin of the employee? If so, protection must be provided for those exposed parts of the body.

Finally, the selection of the respirator to be used must be based on approval standards. The National Institute for Occupational Safety and Health (NIOSH) and the Mine Safety and Health Administration (MSHA) test and certify respirators. Not all respirators being manufactured have been tested or approved by these bodies. In addition, the approval for the respirator may be limited to exposures occurring within a given environmental situation. The concentration of the substance as well as the duration of use must also be compared to that which is stated for an approved respirator.

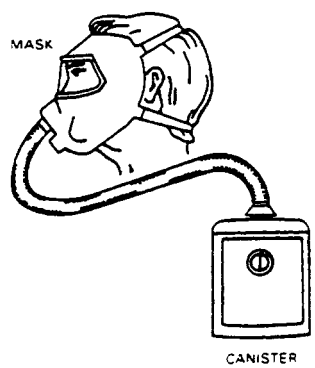
The Air-Purifying Respirator. The first type of respirator that can be used is the air-purifying respirator. The purpose of this respirator is to remove contaminants from the air that is being respired. Sufficient oxygen must be present within the environment to support life. Air-purifying respirators come in three major types: the canister type, the cartridge type, and the mechanical filter type respirator.

The canister-type air-purifying respirator consists of a full facepiece with a hose connecting the mask to a canister. The canister is filled with the material that chemically reacts with the particular known contaminant to render it harmless. The type of materials that can be used to purify air containing contaminants varies with the type of contaminant in the air. Generally, the canister type respirator is limited to certain concentrations of a given substance. These concentrations are stated by the manufacturer in any data sheets involving the respirator and also on the canister itself.

The canisters involve various chemical sorbents for particular materials. For example, canisters exist for removal of acid gases, organic vapors, carbon monoxide, vinyl chloride, ammonia, hydrogen sulfide and chlorine.

Figure 8.4.1

Canister type.



The canisters are color coded to indicate the type of toxic material for which the canister is effective. In general, the color coding is as follows:

- white--acid gases
- black--organic vapors
- green--ammonia
- blue--carbon monoxide
- various color stripes for specific compounds

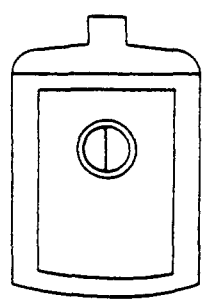
The life of a canister varies and is generally specified based on a standard atmosphere and work rate. Factors that affect the life of a canister are the breathing rate of the individual using the canister, the temperature and humidity of the atmosphere in which the canister is being used, and the concentration of the contaminant in the atmosphere. Obviously each of these variables must be considered when determining how long a particular canister will provide protection for the worker. Standards for the life of canister type respirators have been specified by the Mine Safety and Health Administration. Generally, the typical canister life is approximately thirty minutes, but larger sizes can be purchased when a longer life is desired.

Some canisters come in a window design. The window indicates usage of the canister's capability for purifying the air. The window has a reference half circle of a given color. As the canister is used, the other half circle changes color. When the two colors match, the canister has lost its effectiveness.

Canisters that are used for carbon monoxide generate significant heat. This can provide an indication as to the proper functioning of the canister. If the air is hot when inhaled, the concentration of carbon monoxide in the atmosphere is too high for the canister to be effective, and the worker should immediately exit the area. An additional concern is the fact that the heat that is generated may be dangerous in a flammable atmosphere since it can act as an ignition source for flammable material.

Figure 8.4.2

Canister.



In general, the canister type respirator is for use in emergency situations only. The canister type respirator should not be used for regular work. When the canister is used in an emergency situation, sufficient warning should be given to the individuals so that the respirator can be put to use. The best warning is obtained when the gas has an odor or taste at or near the TLV. However, the odor threshold is most often above a hazardous level. In such cases, once the odor reaches the worker, it is too late to do anything about it. Thus, some type of system should be used to provide a warning when it is necessary for the worker to use a respirator.

After the canister has been used, no matter how long the use has been, it should be discarded. Canister sizes are selected for the appropriate needs of the individual, and a partially used canister can present a potential hazard to the worker unless discarded. The tendency is to assume that the canister is fully operational when in fact, if it has been partially used, it will operate for a shorter period of time to give the appropriate protection.

Cartridge Type Respirators. The cartridge type respirator is essentially a small canister that is attached directly to the facepiece or mask. A half

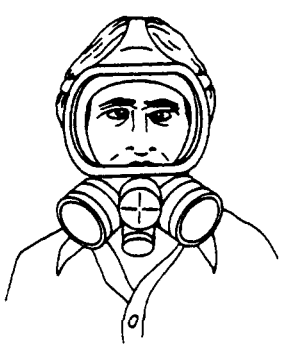
mask with one or two cartridges is most common although a full mask may be used. Cartridge type respirators work in a manner similar to the canister type respirator. The major difference is the size of the smaller cartridge versus that of the canister.

Cartridges are generally approved for use in nonemergency situations only. Such situations involve an exposure to a material that is hazardous only after a prolonged period of time. Cartridge respirators are often used for protection against exposure to certain organic vapors and, in some cases, dust, fumes, and mist.

Mechanical Filter Type Respirator. The mechanical filter type respirator involves a half or full mask with filters attached to the mask in a manner similar to that of the cartridge type respirator. The filters, which are composed of fibrous materials, are connected directly to the mask.

Figure 8.4.3

Cartridge type.



As with any filter, as it begins to clog or fill up, the efficiency is improved. However, as the filter loads, the resistance to breathing is increased and is a limiting factor for the use of the filter.

The filter medium that is used in a mechanical filter type respirator varies. The type of filter used is rated depending upon its efficiency in removing various particle sizes within the atmosphere. These ratings are available for each filter and should be consulted prior to selection.

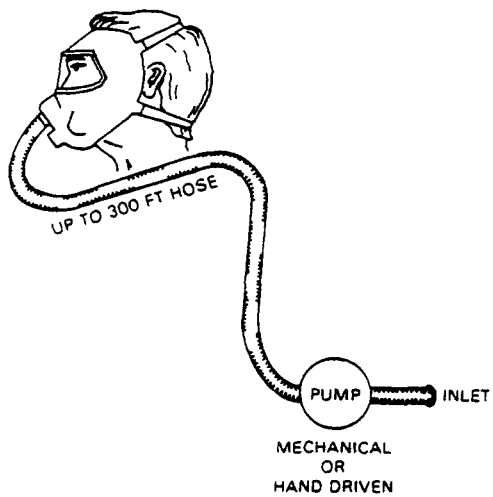
Air-Supplied Respirators. This type of respirator involves a mask to which a supply of air is provided. The air-supplied respirator comes in three general types: the hose mask, the air-line respirator, and the abrasive blasting type respirator.

The hose mask involves a full facepiece with a one-inch collapsible hose attached. The hose is connected to a motor-driven or hand-operated pump outside the comtaminated atmosphere. The hose may also be connected to a

positive air pressure source. The blower intake is located in an uncontaminated area and supplies air from that area to the worker through the hose.

Figure 8.4.4

Hose mask.



This type of system should be of a fail-safe nature to allow for breathing in the event of a failure of the blower. Very little resistance to breathing is encountered using the hose mask, and the worker can use it for a long period of time without fatigue. In general, because of the dynamics of the air within the hose itself, the hose cannot be too long. The maximum length of hose is generally limited to 300 feet.

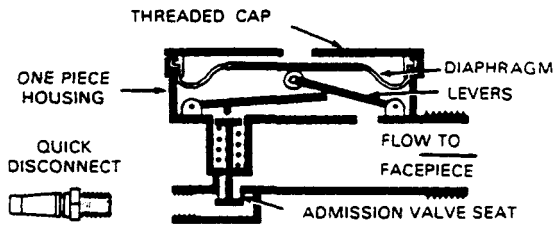
The air-line respirator is essentially the same as the hose mask except it is connected to an air pressure line operating at a volume of between 2 and 20 cubic feet per minute. The connecting line is smaller in diameter and supplies air under pressure. Filtering of compressed air is necessary to assure that clean air is delivered to the worker.

Air-line respirators are often equipped with a demand type valve to regulate the flow of air to the worker. This type of valve opens upon inhalation and closes during exhalation. Exhaled air is discharged through a separate valve.

The air-line respirator usually requires a pressure regulator and a relief valve that is attached to the plant air supply. The pressure regulator lowers the air pressure to that which should be supplied to the worker, and the relief valve acts as a control should the air pressure regulator fail and the pressure become so great as to cause harm to the user.

Figure 8.4.5

Demand valve.



Courtesy of Mine Safety Appliances Co.

The air-line respirator offers low breathing resistance and is suitable for long wear in a contaminated atmosphere.

The abrasive blasting type respirator is composed of either a hose mask or an air-line respirator that is equipped with a helmet, hood, and inner collar that can be worn in an abrasive blasting operation. The hood and helmet protect the worker's face and head from the impact of the abrasive material in the atmosphere.

In an abrasive blasting type respirator, the air enters the hood from the hose and is exhausted at the neck collar or through an exhalation valve.

In certain situations where protection for the entire body is required, an air-supplied suit can be used. As an example, such suits can be used in high heat areas with conditioned air being supplied to the worker.

An additional advantage of the air-supplied respirator or air-supplied suit is that it can be utilized in an area where the oxygen level is such that it will not support life. For reasons of safety, such protection should be provided to the worker when the oxygen level is below 19.5% in the work area.

The Self-Contained Breathing Unit. As is the case with air-supplied respirators, the self-contained breathing unit can be used in areas where there is a deficiency of oxygen. The self-contained breathing unit has the additional advantage of being usable in an area where distance or movement prohibits the use of a hose mask or an air-line respirator.

The self-contained breathing unit is designed to allow the worker to carry a source of respirable air that is attached to the mask. The source of air may be supplied through oxygen-generating chemicals activated by moisture in the user's breath, liquid oxygen in a cylinder, or compressed air in a cylinder. Self-contained breathing units may be designed as either constant-flow type or demand valve operated.

There are two types of self-contained breathing units being manufactured. The first is a recirculating type unit in which a reservoir bag is attached. This bag collects air exhaled from the worker. Oxygen is added to the exhaled air by passing the exhaled air through a canister in which carbon dioxide is retained and oxygen is evolved. The oxygen may be added from compressed or liquid oxygen in a cylinder. This type of mask requires venting since an excess of oxygen is generally produced and must be vented.

The second type is the open circuit type. This type of respirator involves supplying air to the worker directly from a cylinder. The exhaled air is vented to the atmosphere from the mask.

Precautions That Should Be Taken When Using Respirators

There are certain precautions that should be taken when respirators are used by the workers. First, the industrial hygiene engineer should be sure that the appropriate equipment is being used. The mask should be selected to provide protection for those materials to which exposure can occur, and the appropriate type of canister or cartridge should be attached. For example, canister and cartridge type respirators are contaminant-specific. That is, if a canister is designed to remove a particular contaminant, it is not effective in removing another contaminant in the atmosphere. Consult "A Guide to Industrial Respiratory Protection," NIOSH Publ. 76-189, June 1976, and the latest edition of the "NIOSH Respirator Decision Logic" for further guidance on the elements of a respirator program and the selection of respirators.

The equipment should be maintained and cleaned on a regular basis. Equipment that is used regularly should be cleaned before and after each use. Equipment that is used in emergency situations only should be inspected and cleaned before and after each use and at least monthly to assure proper working condition in case an emergency arises. Maintenance should involve looking for defects and damage to the equipment. Any dirt that is present should be removed. The cartridge or canister should be checked to determine if it is the proper type. If an air-supplied respirator is being used, the operation of the pump, pressure-relief valve, pressure regulator, and other related equipment should be checked. After use, the mask should be placed in a plastic bag for storage in a clean, cool, dark, dry place.

When a worker is using a respirator inside a contaminated area, a second worker should be stationed outside the area. A lifeline should be attached to the worker in the contaminated area, and the worker outside the contaminated area should be supplied with respirator equipment.

Employees should be properly trained in the use of respirators. Such training should include why respirators are necessary and the necessity for proper handling of the equipment, proper fit, inspection, cleaning, and storage of the respirator. Where a respirator is used in an emergency condition, the workers should be familiar with emergency procedures and should drill using the procedures on a regular basis.

Workers who are required to use respirators should be in good physical condition. This is necessary since the use of a respirator presents an added burden on the worker in performing duties. In the case of a canister or filter type respirator, breathing is difficult and can result in problems for those individuals with cardiovascular disease.

Canisters that have been used should be discarded. When a respirator is needed, it is important that the worker be able to depend on the time of protection that a normal canister will give. A partially used canister will not provide adequate protection. In addition, once a canister has been used, it is subject to loss of effectiveness more quickly than a closed and sealed canister.

Warning should be provided to the worker when exhaustion of the supply of air or canister service life is nearing. This warning should provide the worker with adequate time to exit the contaminated area without harm to health.

New canisters that are purchased should be checked and stored in a dry place. Before storing, the canisters should be weighed and this weight compared to the weight of the canister immediately prior to use. Such weighing provides data on the moisture absorbed, which is an indication of the loss of effectiveness of the canister. If the seal on the canister has been broken, it should not be kept over one year.

Other Personal Protective Equipment

In addition to respirators, other personal protective equipment can be used to protect the worker from health and safety hazards. For protection of the face and eyes, safety glasses, goggles, face shields, and hoods can be provided to the workers.

Safety glasses are used in situations where the impact of particulate matter may result in damage to the eyes. The objective of safety glasses is to act as a barrier to protect the eyes from particulates. Safety glasses should be equipped with side shields to protect the eye from entry of particulates at the sides of the glasses. Plastic or metal frames can be used. The lenses should be constructed of impact and shock resistant glass. If a significant number of particles are impacting the worker in the facial area, more protection is required.

Goggles protect a larger area of the worker's eyes and can be worn over glasses. The tighter fit offers more protection in the case where more particles are present in the atmosphere.

Where a high incidence of particles are impacting on the face or where there is a potential for minor explosion, the face shield can be used. Face shields offer protection to the entire face, including the eyes. Face shields should cover the side of the head as well as the forehead, face and chin. Safety glasses should be worn underneath the shield for added protection.

In those cases where an extreme hazard exists, the worker can be equipped with a hood covering the face and head as well as the shoulders and upper body. Such a hood is often utilized in welding operations.

In order to protect the worker's head from blows, a hard hat or helmet can be used. These helmets should be constructed in such a manner as to protect the workers not only from the impact of the blow but also should be designed to absorb any shock that might be transferred to the cervical spine. The type of helmet varies and should be chosen for the particular hazards of the job.

As was mentioned earlier, one route of entry for hazardous materials is through the skin. Thus, it is necessary that protective clothing be provided to the worker where the skin is exposed to the systemic action of a toxic material. Where a less toxic material is used in which the physiological result is dermatitis, skin protection is also necessary. Skin protection can also be necessary to protect the worker from radiant heat loads. In this case, reflective type clothing is utilized.

The various types of protective clothing involve gloves, aprons, and full suits. The types of material and design of this clothing are important. The clothing should be chosen in such a way that the fabric does not react to the contaminants. The fabric should also be impervious to the contaminants in order to provide protection from absorption and contact with the skin. The fabric that is involved in such clothing should be flame resistant and of anti-static type. The fabric must also be of sufficient flexibility and design of the clothing such that the worker is able to function with a minimum of retardation to work processes.

Other types of protective equipment can be provided for various types of hazards. Included among these are protection from noise exposure which involves the use of earplugs or earmuffs. The feet can be protected from traumatic injury by using hard-toed shoes, steel sole shoes, instep guards, and in those cases where static electricity is important, conductive soles. In addition, safety showers and eye baths can be provided to flush away any toxic or hazardous substances to which the worker is exposed.

Summary

Personal protective equipment is used as a last resort mechanism for protection of the worker. Other types of controls such as substitution, engineering, and isolation should be instituted before resorting to personal protective equipment. However, there are situations where personal protectiveequipment can and should be used. Where other methods of control are not adequate or in those special cases where an emergency condition arises, the use of personal protective equipment is justified.

Personal protective equipment in the form of respirators can be used to protect the employee from respirable hazards. Three major types of respirators available are: The air-purifying respirator, the air-supplied respirator, and the self-contained breathing unit. Each of these is usable in a particular situation, and certain selection criteria must be utilized when determining which respirator should be used.

Other types of personal protective equipment can be utilized to protect the worker from various potential causes of illness and injury. Among these are protective clothing to protect the skin, eyeglasses and face shields to protect the eyes and face, and earplugs to protect the ears from excessive noise.